

Saurav Bhattarai

Hydrologist / Physically Based Modeling and
Earth Observation

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Research Interests

Physically based hydrological modeling and flood routing; Earth observation of the water balance (precipitation, soil moisture, snow, and water storage); downscaling and bias correction of satellite products; compound hydroclimatic extremes; open research software.

Education

- 2023–2027 **Ph.D. in Engineering**, Jackson State University, Jackson, MS, USA
Department of Civil and Environmental Engineering. Dissertation defense: February 10, 2027; degree expected May 2027.
Dissertation: *Accessing, Refining, and Combining Open Earth Data for a Modular, Parsimonious, Physics-Based Hydrological Model*.
Advisor: Prof. Rocky Talchabhadel. Committee: Dr. Nawa Raj Pradhan (ERDC), Dr. Sadik Khan, Dr. Kejun Wen, Dr. Robert W. Whalin.
- 2023–2025 **M.S. in Engineering (earned en route to the Ph.D.)**, Jackson State University, Jackson, MS, USA, GPA 4.0/4.0
Major: Civil Engineering. Thesis: *Heatwave Dynamics and Urban Cooling: From Risk Assessment to Nature-Based Solutions*.
Advisor: Prof. Rocky Talchabhadel.
- 2017–2022 **Bachelor of Civil Engineering**, Tribhuvan University, Institute of Engineering, Pulchowk Campus, Lalitpur, Nepal, Score 80.14%
- Licensure Fundamentals of Engineering (FE) Civil exam, passed April 2026 (NCEES).

Research Experience

- Aug 2023–Present **Graduate Research Assistant**, Jackson State University, Jackson, MS, USA, Advisor: Prof. Rocky Talchabhadel
- Apply MRRpy without event-wise fitting to floods of the Bagmati River (Kathmandu Valley, Nepal) and the Blue River (Oklahoma, a DMIP benchmark basin), diagnosing error through process representation: after correcting soil hydraulic conductivity with a published global relationship instead of fitting it, MRRpy reproduces eight Bagmati floods with a median Nash–Sutcliffe efficiency of 0.81 with SmartPhones4Water automatic weather stations and 0.54 with raw IMERG rainfall. Also run design floods from return-period rainfall at 15 station locations and a five-year continuous simulation; currently adding baseflow and groundwater recharge.
 - Model the August 2026 Bhote Koshi–Trishuli outburst flood in the Nepal Himalaya as an upstream boundary inflow (preliminary).
 - Led development of the GeoClimate Intelligence Platform [J9, P1], a free, no-code web platform with nine tools over more than 1,000 Earth Engine datasets, used in university coursework and faculty workshops.
 - Detected an abrupt 2017 shift in IMERG V07 rainfall over the Kathmandu Valley with homogeneity tests and long-record gauges, attributed it to lost gauge input, and bias-corrected IMERG with national (DHM) and SmartPhones4Water stations for extreme-rainfall estimation.
 - Downscaled SMAP soil moisture from 9 to 1 km with a transformer and gradient-boosted trees using satellite foundation-model embeddings (AlphaEarth), and evaluated it inside the satellite pixel against airborne campaigns and storm responses.
 - Mapped the timing of the wettest days over global land with ERA5 (1980–2024) [J13]; built a global record of compound heatwave–drought (1950–2025); and, from surge, rainfall, and river records at 79 U.S. tide gauges (1950–2025), attributed the rise in compound flood days to sea-level rise.
 - Completed M.S. research on heat risk and urban cooling [J4, J6, J8, J11]; evaluated satellite precipitation products across the CONUS and Hawaii [J3].

- Oct 2024–Aug 2026 **ORISE Research Fellow**, U.S. Army Engineer Research and Development Center, Coastal and Hydraulics Laboratory, Vicksburg, MS, USA, Mentor: Dr. Nawa Raj Pradhan
- Developed MRRpy, an open-source, distributed, physically based rainfall–runoff and flood-routing model with interchangeable runoff generation (Green–Ampt, variable source area, SCS curve number, impervious surfaces) and routing (kinematic, Muskingum–Cunge, semi-implicit diffusive), Google Earth Engine data retrieval from a basin outlet, and CPU/GPU backends.
 - Built and tested a GSSHA model of the Kathmandu Valley (Bagmati River basin) as an established physically based reference.
 - Implemented the ERDC GLOBAL-SAM snow-depth and SERVES root-zone soil-moisture algorithms as Google Earth Engine applications, and evaluated SERVES against SMAP and 2,643 in situ stations worldwide (ISMN, SCAN, SNOTEL, USCRN, NEON), mapping where it holds and where it does not across climates and seasons.
 - Developed an explainable ensemble of decision trees and neural networks (multilayer perceptrons) that downscales GRACE terrestrial water storage from 55 to 5 km over the Mississippi River Basin, validated against USGS aquifer storage change and 825 groundwater wells.
- May–Aug 2024 **Summer Research Intern**, U.S. Army Engineer Research and Development Center, Vicksburg, MS, USA
- Coupled ADCIRC storm-surge output to GSSHA as boundary conditions, simulated compound rainfall–surge flooding, and estimated joint probabilities with copulas.

Publications

Name in bold. First-author journal articles: [J3], [J4], [J6], [J8], [J9], [J11], [J13]. Google Scholar (September 2026): 130 citations, h-index 7.

Peer-Reviewed Journal Articles

- [J14] Gautam, A., **Bhattarai, S.**, Pradhan, N. R., & Talchabhadel, R. (2026). Urban morphology controls thermal and moisture anomalies: Insights from local climate zones across U.S. cities. *Remote Sensing Applications: Society and Environment*, 43, 102161. doi:10.1016/j.rsase.2026.102161
- [J13] **Bhattarai, S.**, Pradhan, N. R., & Talchabhadel, R. (2026). Do the wettest days occur together? A global analysis on disentangling precipitation intensity from seasonal timing. *Earth and Space Science*, 13(4), e2025EA004845. doi:10.1029/2025EA004845
- [J12] Poudel, S., Bista, S., **Bhattarai, S.**, Sharma, S., & Talchabhadel, R. (2026). Understanding multi-hazard risk for U.S. coastal cities. *International Journal of Disaster Risk Reduction*, 134, 106023. doi:10.1016/j.ijdrr.2026.106023
- [J11] **Bhattarai, S.**, He, C., Sharma, S., Kumar, S., & Talchabhadel, R. (2026). Rainfall relief or humidity havoc? The boon and curse of precipitation during heatwaves. *Natural Hazards*, 122(5), 172. doi:10.1007/s11069-025-07742-x
- [J10] Bista, S., **Bhattarai, S.**, Sharma, S., & Talchabhadel, R. (2026). Local-scale flood hazard projections in historically vulnerable communities. *Modeling Earth Systems and Environment*, 12(1), 5. doi:10.1007/s40808-025-02599-2
- [J9] **Bhattarai, S.**, Pradhan, N. R., & Talchabhadel, R. (2026). GeoClimate Intelligence Platform: A web-based framework for environmental data analysis. *Environmental Modelling & Software*, 197, 106826. doi:10.1016/j.envsoft.2025.106826
- [J8] **Bhattarai, S.**, Banjara, P., Pandey, V. P., Aryal, A., Pradhan, P., Al-Douri, F., Pradhan, N. R., & Talchabhadel, R. (2025). Quantifying the cooling effects of blue-green spaces across urban landscapes: A case study of Kathmandu Valley, Nepal. *Urban Climate*, 61, 102493. doi:10.1016/j.uclim.2025.102493
- [J7] Talchabhadel, R., Panthi, J., Pandey, V. P., Rakhal, B., Ghimire, G. R., Bista, S., **Bhattarai, S.**, Poudel, S., Bhattarai, Y., Prajapati, R., Thapa, B. R., Nepal, B., & Sharma, S. (2025). Yesterday's extremes, today's new normal: Flood risk in the Kathmandu Valley, Nepal. *Natural Hazards*, 121(16), 19409–19423. doi:10.1007/s11069-025-07524-5
- [J6] **Bhattarai, S.**, Bokati, L., Sharma, S., & Talchabhadel, R. (2025). Understanding spatiotemporal variation of heatwave projections across US cities. *Scientific Reports*, 15, 10643. doi:10.1038/s41598-025-95097-5
- [J5] Talchabhadel, R., **Bhattarai, S.**, & Bista, S. (2025). Projected changes in precipitation extremes across the Mississippi River Basin using the NASA global daily downscaled datasets NEX-GDDP-CMIP6. *International Journal of Climatology*, 45(5), e8748. doi:10.1002/joc.8748
- [J4] **Bhattarai, S.**, Bista, S., Sharma, S., White, L. D., Amini, F., & Talchabhadel, R. (2024). Spatiotemporal characterization of heatwave exposure across historically vulnerable communities. *Scientific Reports*, 14, 20882. doi:10.1038/s41598-024-71704-9
- [J3] **Bhattarai, S.**, & Talchabhadel, R. (2024). Comparative analysis of satellite-based precipitation data across the CONUS and Hawaii: Identifying optimal satellite performance. *Remote Sensing*, 16(16), 3058. doi:10.3390/rs16163058
- [J2] Banjara, P., **Bhattarai, S.**, Pandey, V. P., & Talchabhadel, R. (2024). Spatiotemporal characterization of heatwaves on an urban center using satellite-based estimates. *Theoretical and Applied Climatology*, 155(7), 6677–6695. doi:10.1007/s00704-024-05026-1

- [J1] Bhattarai, T. N., Ghimire, S., Aryal, S., Baaniya, Y., **Bhattarai, S.**, Sharma, S., Bhattarai, P. K., & Pandey, V. P. (2023). Projected changes in hydro-climatic extremes with CMIP6 climate model outputs: A case of rain-fed river systems in Western Nepal. *Stochastic Environmental Research and Risk Assessment*, 37(3), 965–987. doi:10.1007/s00477-022-02312-0

Peer-Reviewed Conference Proceedings

- [P1] **Bhattarai, S.**, Pradhan, N. R., Walters, W. L., & Talchabhadel, R. (2026). Bridging the gap: An open-source, low-code platform for integrating data science into water resources engineering education. In *World Environmental and Water Resources Congress 2026* (pp. 80–91). ASCE. doi:10.1061/9780784486931.007

Manuscripts Under Review and in Preparation

- Under review Kalonji, A., Khalil, S., **Bhattarai, S.**, Talchabhadel, R., & Sharma, S. Multiscale evaluation of IMERG satellite precipitation estimates across the United States.
- In preparation **Bhattarai, S.**, Pradhan, N. R., & Talchabhadel, R. MRRpy: A modular Python model for distributed rainfall–runoff generation and flood routing. To be submitted to *Journal of Open Source Software*.
- In preparation **Bhattarai, S.**, Pradhan, N. R., & Talchabhadel, R. Explainable tree–neural network ensemble for GRACE-based terrestrial water storage and groundwater estimation.
- In preparation **Bhattarai, S.**, Talchabhadel, R., & Pradhan, N. R. An abrupt 2017 shift in IMERG V07 rainfall over the Kathmandu Valley: Detection with long-record gauges, attribution to lost gauge input, and consequences for bias correction and extreme-rainfall estimation.
- In preparation **Bhattarai, S.**, Pradhan, N. R., & Talchabhadel, R. A 76-year global record reveals accelerating compound heatwave–drought hazards in drylands.
- In preparation **Bhattarai, S.**, Pradhan, N. R., & Talchabhadel, R. Sea-level rise, not changing storm dependence, explains the rise in compound flood days along the contiguous U.S. coast. Target: *Earth's Future*.

Dataset

- [D1] Talchabhadel, R., & **Bhattarai, S.** (2024). *Interconnected hazards data* [Data set]. DesignSafe-CI. doi:10.17603/ds2-a2vx-sx21

Research Software

- MRRpy Lead developer. Open-source (MIT) distributed rainfall–runoff and flood-routing model, v0.3.0 on PyPI (pip install mrrpy): runs from a basin outlet with Earth Engine data; Python, command-line, and QGIS interfaces; Numba and CuPy acceleration (34× faster on a GPU at 1.38 million cells); an exact linear-time implicit routing solver on the D8 drainage tree, verified against the analytical Hayami solution, with a median mass-balance error of 3 parts per million; automated tests (GitHub Actions) and documentation including an interactive hydrology course. mrrpy.readthedocs.io; github.com/SauravBhattarai19/MRRpy
- GeoClimate Lead developer. GeoClimate Intelligence Platform: no-code Google Earth Engine web platform with nine tools (including GLOBAL-SAM snow depth and SERVES soil moisture) and more than 1,000 datasets; used in university courses and faculty workshops [J9, P1]. geeclimate.streamlit.app; github.com/SauravBhattarai19/GeoClimate-Fetcher
- Earth Engine app Cooling Efficiency Analyzer: estimates the cooling potential of urban green and blue spaces from Landsat land surface temperature [J8]. sauravbhattarai1999.users.earthengine.app/view/cooling-efficiency-analyzer
- Course platform Advanced Water Resources Engineering: interactive Streamlit course materials. advancedwater.streamlit.app

Fellowships and Grants

- 2026 **NASA FINESST** (Future Investigators in NASA Earth and Space Science and Technology), submitted, pending. “Satellite-Derived Antecedent Soil Moisture Deficit for Calibration-Free Initialization of a Gridded, Physics-Based Flood Model.” Future Investigator; conceived the project and wrote the proposal (\$50,000 requested).
- 2024–2026 **ORISE Research Fellowship**, U.S. Department of Defense Research Participation Program, ERDC Coastal and Hydraulics Laboratory; extended based on research performance.
- 2024–2025 **Co-Investigator**, Microsoft–AUC Mini-Grants Program: “Climate Change Impact Assessment Training for Undergraduate Students: Leveraging Planetary-Scale Data and Cloud.”
- 2023–Present **Graduate Research Assistantship and full scholarship**, Hydrological Impacts Computing, Outreach, and Resiliency Partnership (HICORPS), Jackson State University, with ERDC and Woolpert.

Honors and Awards

- 2026 Academic Excellence Award, International Students Association, Jackson State University.
- 2026 National finalist (team leader), Mastercard IGS × AUC Data Science Initiative Data Challenge (top 26 of 106 teams from 13 HBCUs).
- 2025 Best in Show and first position (lead developer, GeoClimate Intelligence Platform), Microsoft × AUC Data Science Initiative Data Challenge.
- 2025 Honorable Mention (team leader), Mastercard × AUC Data Science Initiative Data Challenge (top 18 of 70+ teams; \$1,000).
- 2024 Winner, AGU Catchment Hydrology Subcommittee Early Career Competition (LSTM rainfall–runoff modeling; \$300).
- 2017–2021 Merit-based full scholarship, Institute of Engineering, Pulchowk Campus (ranked 19th of about 15,000 applicants).

Training and Competitive Programs

- 2026 National Data Platform Education Hub Content Workshop, San Diego Supercomputer Center (selected participant; \$2,000 stipend).
- 2025 CUAHSI-SINTER Snow Measurement Field School, Granby, CO (selected participant; \$600 travel award).
- 2025 NCAR Graduate Visitor Program, Boulder, CO (research residency with NCAR scientists).
- 2025 Tethys Summit, Tampa, FL (travel grant).
- 2024 AGU Chapman Conference on Remote Sensing of the Water Cycle, Honolulu, HI (travel grant).
- 2024 Natural Hazards Center CyberTraining Workshop (travel support).

Teaching Experience

- Fall 2025 **Teaching Assistant**, *Advanced Water Resources Engineering (CIV 465)*, Jackson State University
Designed modules on big-data analytics, machine learning, extreme-value analysis, and climate projections; built the course's interactive platform; delivered lectures and Python/GIS training.
- Spring 2024, 2025 **Teaching Assistant**, *Water Resources Engineering (CIV 370)*, Jackson State University
Taught frequency analysis, unit hydrographs, hydrologic routing, open-channel hydraulics, and stormwater design; designed assignments and exams.
- Spring 2024 **Teaching Assistant**, *Fluid Mechanics Laboratory (CIVL 330)*, Jackson State University
Ran weekly laboratory sessions and graded reports.
- 2022–2023 **Mathematics and Physics Teacher**, *Albright Academy*, Pokhara, Nepal
Variable schedule (about four months in total).

Mentoring

- Howard University (2 undergraduate researchers): evaluation of IMERG precipitation against stations and MRMS; manuscript under review.
- Tribhuvan University, Nepal (5 undergraduate researchers): theses on heatwaves in Nepal.
- Jackson State University (2 undergraduate researchers): development and testing of the GeoClimate Intelligence Platform.
- 2026 Research Mentor, AUC Data Science Initiative PreFreshman Experience (incoming HBCU students; team data-science projects).

Presentations

Oral Presentations (Presenting Author)

- 2026 **Bhattarai, S.**, Pradhan, N. R., Walters, W. L., & Talchabhadel, R. Bridging the gap: An open-source, low-code platform for integrating data science into water resources engineering education. ASCE-EWRI World Environmental and Water Resources Congress, Mobile, AL, April.
- 2026 **Bhattarai, S.**, Pradhan, N. R., & Talchabhadel, R. Democratizing hydroclimatic science education through the GeoClimate Intelligence Platform. 106th AMS Annual Meeting, January.
- 2025 Talchabhadel, R., **Bhattarai, S.**, Bista, S., Poudel, S., Jones, D., & Terrett, S. Democratizing Earth science data for community-based water hazard assessment: A co-production platform for hydrologic intelligence. AGU Fall Meeting, New Orleans, LA, December.

- 2025 **Bhattacharai, S.**, Pradhan, N. R., & Talchabhadel, R. Advancing groundwater monitoring and downscaling using open datasets and machine learning in the Mississippi River Basin. XIX World Water Congress, Marrakech, Morocco, December.
- 2025 **Bhattacharai, S.**, & Talchabhadel, R. Evaluating the compound effects of precipitation and storm surge on coastal flooding risk in the CONUS. ASCE-EWRI World Environmental and Water Resources Congress, Anchorage, AK, May.
- 2024 **Bhattacharai, S.**, & Talchabhadel, R. Assessing cascading and compound climate extremes and associated vulnerabilities across the United States. 3rd International Conference on Natural Hazards and Risks in a Changing World, Amsterdam, Netherlands, June.
- 2024 **Bhattacharai, S.**, Sharma, S., & Talchabhadel, R. Assessing multidimensional climate extremes and associated vulnerabilities across the United States. EGU General Assembly, Vienna, Austria, April (EGU24-20600).
- 2023 **Bhattacharai, S.**, Bokati, L., Sharma, S., & Talchabhadel, R. Assessing heatwave impacts and enhancing climate resilience in highly populated US cities. AGU Fall Meeting, San Francisco, CA, December.

Poster Presentations (Presenting Author)

- 2026 **Bhattacharai, S.**, Pradhan, N. R., & Talchabhadel, R. Explainable tree-neural network ensemble for GRACE-based TWS downscaling and groundwater estimation. ERDC RD26 Workshop, Vicksburg, MS.
- 2025 Talchabhadel, R., **Bhattacharai, S.**, Prajapati, A. K., Karn, B., Bhattacharai, B., Yadav, R., & Bhattacharai, P. K. Spatiotemporal assessment of urban heat island and heatwave dynamics in densely populated Nepalese cities: Evaluating nature-based cooling strategies. AGU Fall Meeting, New Orleans, LA, December.
- 2025 **Bhattacharai, S.**, Talchabhadel, R., & Pradhan, N. R. Precipitation temporal disaggregation for event-based hydrological modeling in the Kathmandu watershed, Nepal. AGU Fall Meeting, New Orleans, LA, December.
- 2024 **Bhattacharai, S.**, Byrd, A., & Talchabhadel, R. Evaluating the compound effects of precipitation and storm surge on coastal flooding risk in the CONUS: A detailed case study of Galveston. AGU Fall Meeting, Washington, DC, December.
- 2024 **Bhattacharai, S.**, Sharma, S., & Talchabhadel, R. Detecting multihazards in the United States using machine learning: A comparative analysis of individual and compound hazard events. AGU Fall Meeting, Washington, DC, December.
- 2024 Talchabhadel, R., **Bhattacharai, S.**, Banjara, P., & Pandey, V. P. Urban cooling strategies: Evaluating the effectiveness of green spaces and water bodies in mitigating heat exposure in Kathmandu Valley, Nepal (GC51H-0059). AGU Fall Meeting, Washington, DC, December.
- 2024 **Bhattacharai, S.**, & Talchabhadel, R. Heatwaves, humidity, and the water cycle: A remote sensing perspective on climate transformation. AGU Chapman Conference on Remote Sensing of the Water Cycle, Honolulu, HI, February.

Co-Authored Abstracts (Presented by Co-Authors)

- 2025 Pradhan, N. R., **Bhattacharai, S.**, Kloss, G., et al. Global Snow Attribute Mapping Tool (Global-SAM). AGU Fall Meeting, New Orleans, LA. [Oral]
- 2025 Gautam, A., **Bhattacharai, S.**, & Talchabhadel, R. Unraveling urban heat and moisture anomalies across the continental U.S. using local climate zones and Earth observation data. AGU Fall Meeting, New Orleans, LA. [Poster]
- 2025 Dahal, B., **Bhattacharai, S.**, Bhattacharai, P. K., & Talchabhadel, R. Evaluating the applicability of SWOT and SWORD for Nepal's water monitoring: Building a scalable national database. AGU Fall Meeting, New Orleans, LA. [Poster]
- 2025 Talchabhadel, R., **Bhattacharai, S.**, Bista, S., Poudel, S., & Kafoury, R. Projecting extreme temperatures and precipitation for the Gulf Coast region of the United States. ASCE-EWRI World Environmental and Water Resources Congress, Anchorage, AK. [Oral]
- 2024 Talchabhadel, R., Bernard, D., Johnson, K., **Bhattacharai, S.**, Bista, S., Poudel, S., & Kafoury, R. Evaluation of climate change projections using open-source software infrastructure across the Gulf Coast of the United States (NH13B-2293). AGU Fall Meeting, Washington, DC. [Poster]
- 2024 Bista, S., **Bhattacharai, S.**, Sharma, S., & Talchabhadel, R. Implications of non-stationary climate dynamics for flood risk projections (H05-13). AGU Fall Meeting, Washington, DC. [Oral]
- 2024 Banjara, P., **Bhattacharai, S.**, Pandey, V. P., & Talchabhadel, R. Assessing heatwave dynamics and vulnerability in Nepal: A geospatial analysis (H34F-04). AGU Fall Meeting, Washington, DC. [Oral]
- 2024 Bista, S., Bhattacharai, Y., **Bhattacharai, S.**, Sharma, S., Talchabhadel, R., & Amini, F. Urban flood inundation prediction under climate change. ASCE-EWRI World Environmental and Water Resources Congress, Milwaukee, WI. [Oral]
- 2024 Talchabhadel, R., **Bhattacharai, S.**, Bista, S., White, L., & Amini, F. Providing undergraduate students with training on evaluating climate change projections within the framework of open-source software infrastructure. ASCE-EWRI World Environmental and Water Resources Congress, Milwaukee, WI. [Oral]

- 2024 Talchabhadel, R., **Bhattarai, S.**, & Bista, S. Empowering undergraduate students in remote sensing for climate change projection evaluation using open-source software infrastructure. AGU Chapman Conference, Honolulu, HI. [Poster]
- 2024 Talchabhadel, R., **Bhattarai, S.**, Banjara, P., & Sharma, S. Projecting heatwave consequences in urban centers of the United States under changing climate. AMS 15th Conference on Environment and Health, Baltimore, MD. [Oral]
- 2024 Thapa, B. R., Talchabhadel, R., & **Bhattarai, S.** Translating multi-hazard risk assessment to compounding and cascading disaster risk management for resilient infrastructure. 2nd GeoMandu International Conference, Kathmandu, Nepal, November. [Invited]

Workshops Taught and Outreach

- 2025 Lead Instructor, Smartphones4water Workshop: GeoClimate Intelligence Platform for crowdsourced and accessible environmental data analysis.
- 2025 Co-Instructor, AUC Data Science Initiative Faculty Workshop, "Harnessing GeoClimate Intelligence for Research Innovation" (28 registered faculty).
- 2025 Lead Instructor, Python Training Workshop, Jackson State University: hydroclimate and geospatial applications.
- 2025 Program Manager and Facilitator, Mastercard × AUC Data Science Initiative Regional Pitch Stop, Jackson State University.

Professional Service

- Peer review *Communications Earth & Environment* (2), *Earth and Space Science* (1), *Urban Climate* (1).
- Convening Student Convener, AGU Fall Meeting 2024 session NH13B, Coastal Storm Risk Modeling and Analysis via Cloud Computing.
- Memberships American Geophysical Union; European Geosciences Union; American Society of Civil Engineers; Nepal Engineering Council.

Technical Skills

- Modeling MRRpy (developer); GSSHA (WMS); ADCIRC (mesh generation in SMS), coupled to GSSHA; 1D and 2D finite-volume shallow-water solvers (implemented, dam-break problems); HEC-HMS and HEC-RAS (basic).
- Machine learning Tree ensembles (XGBoost, LightGBM, CatBoost), multilayer perceptrons, LSTM, and transformers (scikit-learn, PyTorch); satellite foundation-model embeddings (AlphaEarth); SHAP interpretation; spatial cross-validation.
- Programming Python (NumPy, SciPy, pandas, xarray, rioarray, Dask, Zarr, GeoPandas, rasterio); Numba and CuPy (GPU); Google Earth Engine (Python and JavaScript); Streamlit; working knowledge of HTML, CSS, and JavaScript.
- Software practice Git and GitHub; pytest; GitHub Actions continuous integration; PyPI packaging; MkDocs and Read the Docs; QGIS plugin development; Linux and HPC batch processing.
- Data *Satellite*: GRACE/GRACE-FO, SMAP, IMERG, MODIS, Landsat. *Reanalysis and climate*: ERA5, GLDAS, DAYMET, NEX-GDDP-CMIP6. *Land surface*: SoilGrids, HiHydroSoil, FABDEM, ESA WorldCover. *In situ*: ISMN, SCAN/SNOTEL, USCRN, NEON, USGS streamflow and wells, NOAA tide gauges, DHM and S4W stations (Nepal).
- Statistics Extreme-value and frequency analysis; copulas; quantile mapping and bias correction; homogeneity tests (Pettitt, SNHT, Buishand).

References

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